MD ADNAN ROSUL

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EDUCATION

Master of Science – Computer Science and Engineering 2026 – Present

Major: Data Science

Brac University

Bachelor of Engineering – Electronics-Computer & Information Engineering 2017 – 2021

International Islamic University Malaysia (IIUM)

Final CGPA: 3.06 / 4.00 | Dean's List Award (2020)

Thesis: White-Box Image Cartoonization via Generative Adversarial Networks

Thesis Supervisor: Assoc. Prof. Dr. Nurul Fariza Zulkurnain

Relevant Coursework: Data Structures & Algorithm Design, Data Mining, Computer Architecture & Systems Design, Digital Signal Processing, Computational Methods & Statistics, Database Systems, Computer Communications & Networks, Operating Systems, Differential Equations

UNDERGRADUATE THESIS RESEARCH

White-Box Image Cartoonization Using Generative Adversarial Networks — IIUM 2021

- Developed a white-box GAN for image cartoonization using surface, structure, and texture representations. Curated 4,000 images (2,000 photographs and 2,000 cartoons) at 256 × 256 resolution.
- Combined adversarial, content-preservation, and structural-reconstruction losses; trained with Adam for 10,000 iterations (batch size 25) on Google Colab Pro, using approximately 13.5 GB VRAM over 9–10 hours.
- Tracked Fréchet Inception Distance (FID) every 500 iterations. Qualitative comparisons showed clearer boundaries, fewer color artifacts, and better fine-detail preservation.
- Built a preprocessing pipeline with guided bilateral filtering and Felzenszwalb superpixel segmentation; used CycleGAN-based unpaired image translation to work without paired training data.
- Reviewed 15+ generative modeling frameworks, including CartoonGAN, CariGAN, and Conditional WGAN, comparing loss functions, evaluation metrics, and dataset strategies.

OTHER PROJECTS

IIUM Lazada Mobile E-Commerce Application 2020 – 2021

- Designed a mobile application with backend service integrations, applying database normalization, RESTful API design, and modular frontend architecture.

Smart Mosquito Repellent — Autonomous IoT Sensor System 2020

- Designed and built a hardware system integrating sensor-triggered actuator control with embedded processing logic, demonstrating hardware-software co-design and real-time embedded system development.

TECHNICAL SKILLS

- Systems: TensorFlow model training; VRAM-aware batch scheduling; cloud checkpointing; REST APIs; IoT monitoring; real-time KPI dashboards.
- AI & Computer Vision: White-box GANs, CycleGAN, Conditional WGAN, CNNs, VGG feature extraction, sparse coding, morphological component analysis, patch discriminators.
- Data Engineering: SQL query optimization, Power BI, Python automation, dataset curation and quality control.
- Programming & Libraries: Python, C, C++, JavaScript; TensorFlow, NumPy, OpenCV, Matplotlib.

RELEVANT WORK EXPERIENCE

Senior Systems Engineer — VX Services Limited | Bangladesh Jan 2023 – Feb 2025

- Built SQL analytical pipelines and Power BI dashboards to monitor Grameenphone's nationwide CCTV infrastructure, supporting 99.9% uptime across distributed sensor nodes.
- Deployed REST APIs for attendance analytics and IoT fire-detection monitoring, aggregating and synchronizing data across reporting systems.
- Led technical reporting and performance reviews, translating system telemetry into analytical reports for executives and cross-functional stakeholders.

Systems Engineer — VX Services Limited | Bangladesh May 2022 – Dec 2022

- Maintained Paxton Net2 and HikCentral Professional access-control and surveillance systems across multiple sites, using systematic debugging and fault isolation.
- Built executive dashboards combining live sensor data through data normalization and multi-source aggregation.

Software & IT Intern — Learnio | Remote, Denmark Nov 2021 – Jan 2022

- Performed QA testing and contributed UI/UX improvements to educational software platforms during a remote internship.

RESEARCH INTERESTS

- Efficient & Scalable AI Systems
- Generative AI Models
- Explainable & Secure AI
- Scalable & Reliable ML Systems

CERTIFICATIONS, AWARDS & LEADERSHIP

Aspire Leaders Program — Harvard Leadership Initiative 2025
Dean's List Award — International Islamic University Malaysia 2020
Fundamentals of Digital Marketing — Google Digital Garage 2020
8th Place — BAS Divisional Science Olympiad — Bangladesh Academy of Sciences 2013
JSC Result Scholarship — National Board of Bangladesh 2011

Leadership & Volunteering

- Core Committee Member — IIUM Bangladesh Community 2018 – 2019
  Coordinated academic and cultural programs for the international student community.
- Volunteer, Youth Sport Leadership Programme — British Council 2011
  Supported youth development and community engagement activities.